

AUTOMATIC SCHOOL BELL

With Adjustable Timings

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This proposed automatic bell is an electronic system that rings school bells at predefined times without manual intervention. It is built around an Arduino Uno, which is programmed to control a buzzer or electric bell based on a preset schedule, eliminating the need for manual ringing.

An RTC (real-time clock) module ensures precise timekeeping, allowing administrators to set and adjust bell timings as required. An

LCD display can be added to show the current time and upcoming bell schedules. This system improves efficiency, maintains a disciplined routine, and removes dependency on manual operation.

Table 1 lists the components required to build this device, while Fig. 1 shows the authors' prototype.

Circuit and working

Fig. 2 shows the circuit diagram of the automatic school bell. It is built around Arduino Uno R3, 16x2 LCD display with I2C (MOD1), DS3231 RTC, 4x4 matrix keypad, 5V SPDT relay, 230V AC call bell, and a few other essential components. A 9V, 1A adaptor is used to supply power to the Arduino Uno after uploading the source code.

The system features a built-in real-time clock (DS3231), which tracks real-time data. When the current time matches the preset bell ringing time, the relay energises and switches on the bell. The bell

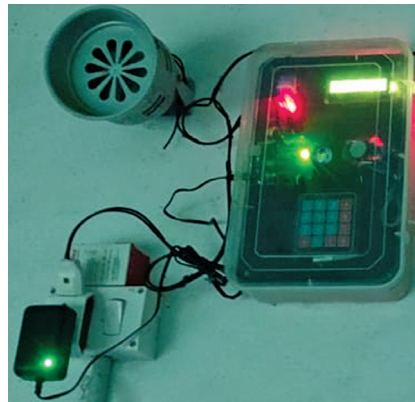


Fig. 1: Authors' prototype

TABLE 1 BILL OF MATERIALS	
Components	Quantity
Arduino Uno R3 (MOD1)	1
DS3231 RTC	1
16x2 LCD display with I2C (MOD2)	1
4x4 matrix keypad	1
5V SPDT relay	1
9V, 1A adaptor for power supply to the Arduino board	1
230V AC call bell	1
Buzzer	1
Jumper wires and other wires	As required

timings can be modified at any time using the 4x4 keypad, making it adaptable for regular school schedules as well as examination periods.

The real-time clock is displayed on a 16x2 LCD display, providing a clear view of the day, date, and time. The Arduino Uno controls all functions, receiving input via the keypad and storing it in its EEPROM memory. When the stored bell time

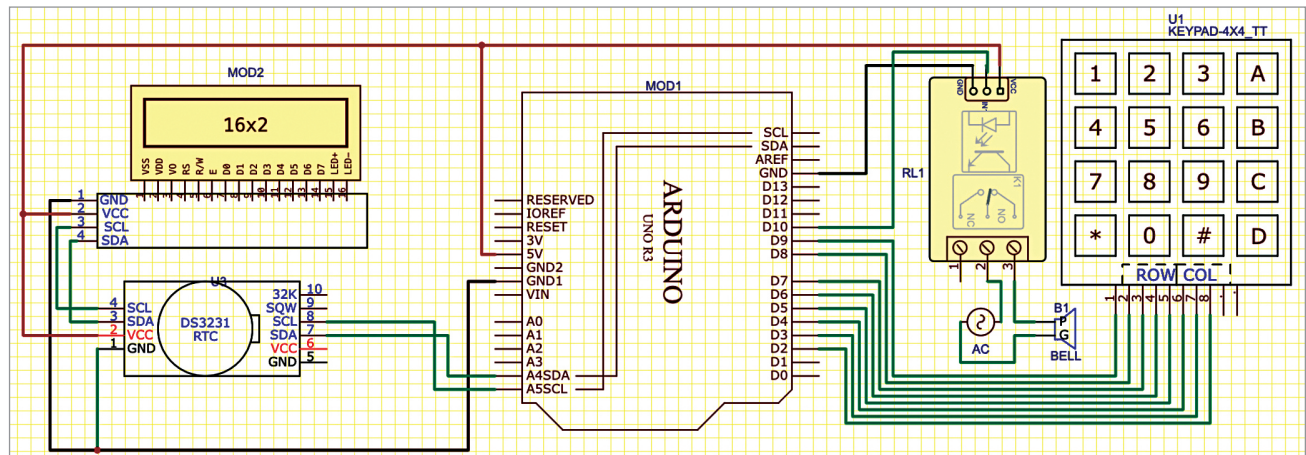


Fig. 2: Circuit diagram